

Semantic Contract & AI Quality Controls

3B. SEMANTIC CONTRACT SPECIFICATION

- Core Datasets & Grains:**
 - claims_ledger.csv: Submitted medical claim transaction line items. Grain: Exactly one claim event per row (claim_id).
 - patient_accounts.csv: Patient coverage and eligibility master data. Grain: One row per patient/member account.
 - procedure_catalog.csv: Clinical CPT/HCPCS codes and allowed rates. Grain: One row per procedure code.
- Units of Measure:** Monetary amounts (billed_amount, allowed_amount) in USD; length_of_stay and days_between_service_and_claim in integer days; patient_age in years.
- Point-in-Time / As-Of Meaning:** Claim evaluation relies strictly on state as-of date_of_service (HT-FAQ-001 Q2). Member eligibility must be joined temporally (effective_start_date ≤ date_of_service ≤ effective_end_date). Emergency policy windows (e.g., HT-MEMO-001 copay waiver between Feb 1 – Apr 30, 2026) apply dynamically.
- Source of Truth & Freshness:** Claims Adjudication Engine and Fee Schedule Matrix (HT-GUIDE-002). Clean claims adjudicated within 30 days (HT-REG-001); feature store refreshed T+1.
- Contract/Version & Interpretation:** Version v2.0. Enforces NCCI unbundling (HT-FAQ-001 Q3 / procedure_catalog.csv) and senior review routing for claims >\$25,000 (HT-PROC-001).

REQUIRED SEMANTIC-FAILURE EXAMPLE

1. Schema-Valid but Semantically Wrong Record:

```
{
  "claim_id": "C00000088",
  "member_id": "MBR-00042",
  "procedure_code": "99285",
  "billed_amount": 1200.00,
  "allowed_amount": 1200.00,
  "date_of_service": "2026-02-15",
  "decision_status": "Approved"
}
```

2. Explanation of AI Misinterpretation:

The payload satisfies structural schema constraints (allowed_amount ≤ billed_amount, valid data types). However, the engine calculates allowed_amount using present-day fee schedules (\$1,200.00) rather than the rate effective on date_of_service (\$850.00). Additionally, it joins member eligibility using present-day status ("Active") despite coverage being inactive on the service date, and fails to apply regional copay waiver HT-MEMO-001. Downstream AI models accept this record as valid baseline reimbursement, underestimating financial risk and violating prompt pay compliance (HT-REG-001).

3. Corrected / Canonical Representation:

```
{
  "claim_id": "C00000088",
  "member_id": "MBR-00042",
  "procedure_code": "99285",
  "billed_amount": 1200.00,
  "allowed_amount": 0.00,
  "date_of_service": "2026-02-15",
  "decision_status": "Denied",
  "denial_reason_code": "ERR_ELIG_POINT_IN_TIME"
}
```

AI CONTEXT / QUALITY CONTROLS TABLE

RULE / EXPECTATION	WHAT GOES WRONG	WHAT AI RECEIVES	POSSIBLE AI IMPACT	CONTROL / RESPONSE
Monetary Unit Consistency (billed_amount, allowed_amount)	Ingestion pipeline ingests source data formatted in thousands instead of single USD decimals.	billed_amount = 120 instead of 120000	Risk and utilization models severely underestimate claim cost magnitude and distort financial severity features.	Automated schema normalization check + assertion gate verifying values fall within expected standard deviation bounds.
Point-in-Time Fee Schedule Validity	Source system calculates reimbursement using	allowed_amount reflects current fee	Audit models falsely validate decision	Point-in-time join assertion in feature engineering ETL

RULE / EXPECTATION	WHAT GOES WRONG	WHAT AI RECEIVES	POSSIBLE AI IMPACT	CONTROL / RESPONSE
(allowed_amount as-of date_of_service)	current fee schedules instead of the policy rate effective on date_of_service.	schedule (\$1,200.00) rather than 2026-02-15 rate (\$850.00)	compliance, leading to unnoticed overpayment leakage and policy drift.	to enforce temporal alignment with effective policy tables.
Temporal Coverage Boundary (patient_accounts Eligibility Join)	Pipeline joins member coverage using present-day eligibility_status rather than status active on date_of_service.	eligibility_status = Active (current) for a claim incurred during a historical coverage gap	Automated claims adjudication models predict approval for non-covered events, resulting in improper billing.	Temporal feature store control enforcing bi-temporal range joins (effective_start_date ≤ date_of_service ≤ effective_end_date).

MACHINE-READABLE JSON SEMANTIC CONTRACT (SEMANTIC_CONTRACT_V2.JSON)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "contract_name": "Healthcare AI Microcosm Semantic Contract",
  "version": "v2.0",
  "target_domain": "Healthcare Claims Adjudication & Risk Prediction",
  "core_datasets": {
    "claims_ledger": {
      "description": "Submitted healthcare claim line items and adjudication decisions.",
      "grain": "One row per discrete claim transaction event (claim_id).",
      "business_rules": [
        "billed_amount must be >= allowed_amount.",
        "allowed_amount is valid strictly as-of date_of_service based on fee schedule in effect.",
        "decision_status relies on point-in-time member eligibility on date_of_service, not current ingest date."
      ]
    },
    "patient_accounts": {
      "description": "Patient demographic, plan coverage, and temporal eligibility records.",
      "grain": "One row per patient/member account.",
      "business_rules": [
        "eligibility_status must be evaluated bi-temporally (effective_start_date <= date_of_service <= effective_end_date).",
        "plan_code links directly to claims_ledger coverage rules."
      ]
    },
    "procedure_catalog": {
      "description": "Standardized procedure codes, clinical complexity, and fee schedules.",
      "grain": "One row per procedure code (CPT/HCPCS).",
      "business_rules": [
        "Codes with review_flag = 'Y' require NCCI unbundling validation (modifier -59).",
        "Establishes maximum allowed reimbursement bounds per procedure."
      ]
    }
  },
  "metadata_policy_rules": [
    {
      "rule_id": "HT-FAQ-001_Q2",
      "type": "Temporal Boundary",
      "description": "Member eligibility joins must be executed point-in-time as of date_of_service."
    },
    {
      "rule_id": "HT-MEMO-001",
      "type": "Regional Policy Waiver",
      "description": "Emergency ER visits (CPT 99285) in US-SE (FL) between Feb 1 - Apr 30, 2026 require copay waiver ($0.00).",
      "waiver": "$0.00"
    },
    {
      "rule_id": "HT-REG-001",
      "type": "Prompt Pay Compliance",
      "description": "Clean claims must be adjudicated within 30 calendar days."
    }
  ]
}
```